

Pure-state homogeneity and shell matching

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Scope

Two-sided inner sequences establish pure-state homogeneity, which supplies one fixed shell family. Each non-root component uses the same approximately inner automorphism for all its shells; the generic matching theorem retains its stated hypotheses.

8 direct Cards + 15 reused prerequisites = 23 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Pure-state homogeneity of the completed CAR algebra
- Key construction: Choosing the CAR shell family from proved homogeneity

[Read this route with prerequisites](#)

Detailed route conditions

For pure states φ, ψ on CAR, the sequence construction controls both $\alpha_n(a)$ and the actual inverses $\alpha_n^{-1}(a)$. The limiting automorphism satisfies $\varphi \circ \alpha = \psi$. The detailed alternating correction inequalities, the protected inverse images, and the substitution $R_n^{-1}(d_j)$ are retained in the key proof Card; this route does not duplicate that proof.

For one fixed approximately inner automorphism, shell matching gives $w_{i,n}^* w_{i,n} = \alpha_i(p_n)$ and $w_{i,n} w_{i,n}^* = p_n$, where $p_n = f_n - f_{n+1}$. At the root class $o = [\omega]$, the chosen family uses $\alpha_o = \text{id}$ and $w_{o,n} = p_n$. Each other component uses one automorphism for all of its shells. The generic projection-matching theorem is not a new proof of the general KOS proposition. The same chosen homogeneity family is used in the construction of the algebra and in the classification of its irreducible representations.

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (3 Cards)

Level 0 Exact projection links for one approximately inner automorphism A point-norm approximation followed by close-projection conjugacy gives exact support identities for an arbitrary projection family. MathlibAnnex.CStarAlgebra.exists_shell_family_of_approximately_inner
Immediate Card prerequisites: None in this selected scope Used by in this scope: Choosing the CAR shell family from proved homogeneity
Open Declaration Card Read exact source Level 0 A fixed representative family of CAR shell data One family records a transporting automorphism and exact shell links for each selected pure-state class, with an explicitly fixed root component. MathlibAnnex.CStarAlgebra.CAR.RepresentativeShellFamily
Immediate Card prerequisites: None in this selected scope Used by in this scope: Choosing the CAR shell family from proved homogeneity
Open Declaration Card Read exact source Level 0 The CAR algebra as a completion of finite matrix stages Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation. MathlibAnnex.CStarAlgebra.CAR.Limit
Immediate Card prerequisites: None in this selected scope Used by in this scope: The product-vector state on the completed CAR algebra · Approximately inner homogeneity of pure CAR states · A finite-row average centralizes its matrix stage · The decreasing root projections in the CAR completion · The completed CAR algebra is infinite-dimensional · A two-sided inner intertwining sequence for two states
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Level 1 (6 Cards)

Level 1

A two-sided inner intertwining sequence for two states

An inner sequence records both forward and actual inverse convergence data before any limiting automorphism is constructed.

`MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence`

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences

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Level 1

Approximately inner homogeneity of pure CAR states

The homogeneity property asks for exact state transport by an automorphism that is locally approximable by inner automorphisms.

`MathlibAnnex.CStarAlgebra.CAR.PureStateHomogeneity`

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: Two-sided inner sequences imply pure-state homogeneity

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Level 1

The product-vector state on the completed CAR algebra

Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion.

`MathlibAnnex.CStarAlgebra.CAR.rootState`

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: Root compression becomes scalar in norm · Purity of the completed root state

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Level 1

The completed CAR algebra is infinite-dimensional

Rules out finite dimension by retaining matrix subalgebras of unbounded dimension.

`MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional`

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: An irreducible CAR representation has no nonzero compact image

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Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. MathlibAnnex.CStarAlgebra.CAR.rootFlag Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm Open Declaration Card Read exact source
Level 1 A finite-row average centralizes its matrix stage Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage. MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path Open Declaration Card Read exact source
Level 2 (3 Cards) Level 2 Two-sided inner sequences imply pure-state homogeneity The forward and inverse limits close the homogeneity argument once an intertwining sequence is supplied for every pure-state pair. MathlibAnnex.CStarAlgebra.CAR.homogeneity_of_innerIntertwiningSequences Immediate Card prerequisites: Approximately inner homogeneity of pure CAR states · A two-sided inner intertwining sequence for two states Used by in this scope: Pure-state homogeneity of the completed CAR algebra Open Declaration Card Read exact source
Level 2 Root compression becomes scalar in norm Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra. MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion Used by in this scope: Simplicity of the completed CAR algebra Open Declaration Card Read exact source

Level 2

Purity of the completed root state

Passes extremality of the root vector states from every finite matrix stage to the completed CAR algebra.

MathlibAnnex.CStarAlgebra.CAR.isPureState_rootState

Immediate Card prerequisites: The product-vector state on the completed CAR algebra

Used by in this scope: Choosing the CAR shell family from proved homogeneity

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Level 3 (1 Card)

Level 3

Simplicity of the completed CAR algebra

Every nonzero closed two-sided ideal of the CAR algebra contains the identity.

MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit

Immediate Card prerequisites: Root compression becomes scalar in norm

Used by in this scope: An irreducible CAR representation has no nonzero compact image

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Level 4 (1 Card)

Level 4

An irreducible CAR representation has no nonzero compact image

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image

Immediate Card prerequisites: Simplicity of the completed CAR algebra · The completed CAR algebra is infinite-dimensional

Used by in this scope: Lifting a corner involution to an ambient CAR unitary · Realizing a finite Gram matrix in a represented CAR corner

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Level 5 (2 Cards)

Level 5

Lifting a corner involution to an ambient CAR unitary

A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential.

`MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path

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Level 5

Realizing a finite Gram matrix in a represented CAR corner

The infinite-dimensional root corner contains an isometric copy of any finite family of vectors.

`MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram`

Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image

Used by in this scope: Purifying a vector state on a finite CAR stage

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Level 6 (2 Cards)

Level 6

Purifying a vector state on a finite CAR stage

Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation.

`MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage`

Immediate Card prerequisites: Realizing a finite Gram matrix in a represented CAR corner

Used by in this scope: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set

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Level 6

Exact vector transport along an almost central unitary path

Finite matrix-state tests ensure exact transport while protecting a prescribed finite set throughout the path.

`MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_exact_unitary_path_apply_eq_and_conjugate_sub_norm_lt`

Immediate Card prerequisites: Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage

Used by in this scope: Cross-representation state approximation with a protected finite set

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Level 7 (2 Cards)

Level 7 Approximating another vector state along a unitary path A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector. MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source
Level 7 Cross-representation state approximation with a protected finite set Finite entrywise tests allow new state approximation while an entire unitary path nearly fixes earlier data. MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Exact vector transport along an almost central unitary path Used by in this scope: Pure CAR states admit two-sided inner intertwining sequences Open Declaration Card Read exact source

Level 8 (1 Card)

Level 8 Pure CAR states admit two-sided inner intertwining sequences Finite-set vector-state transport supplies explicit corrections; their summable conjugation bounds and state estimates yield the required two-sided sequence. MathlibAnnex.CStarAlgebra.CAR.hasInnerIntertwiningSequence_of_pure Immediate Card prerequisites: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set · A two-sided inner intertwining sequence for two states Used by in this scope: Pure-state homogeneity of the completed CAR algebra Open Declaration Card Read exact source

Level 9 (1 Card)

Level 9

Pure-state homogeneity of the completed CAR algebra

The constructed inner intertwining sequences discharge the supplier hypothesis and give an approximately inner transporting automorphism.

`MathlibAnnex.CStarAlgebra.CAR.homogeneity`

Immediate Card prerequisites: Two-sided inner sequences imply pure-state homogeneity · Pure CAR states admit two-sided inner intertwining sequences

Used by in this scope: Choosing the CAR shell family from proved homogeneity

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Level 10 (1 Card)

Level 10

Choosing the CAR shell family from proved homogeneity

CAR homogeneity and exact shell matching supply one family whose distinguished component is fixed by an explicit identity choice.

`MathlibAnnex.CStarAlgebra.CAR.homogeneityShellFamily`

Immediate Card prerequisites: Purity of the completed root state · Exact projection links for one approximately inner automorphism · A fixed representative family of CAR shell data · Pure-state homogeneity of the completed CAR algebra

Used by in this scope: None in this selected scope

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